

Inverted takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy is a transient acute left ventricular dysfunction characterized by left ventricular apical akinesis and ballooning without obstructive coronary disease described predominantly in post-menopausal women in the setting of acute emotional or physical stress. Recent reports have described isolated transient basal akinesis (inverted takotsubo cardiomyopathy) in mostly female patients with acute neurologic disorders or pheochromocytoma. We describe a rare case of a 78-year-old male with inverted takotsubo cardiomyopathy in the setting of acute abdominal pain attributed to biliary colic. A review of published literature reveals that inverted takotsubo cardiomyopathy precipitated by acute stress rather than an acute neurologic disorder appears to be an extremely rare presentation in a male patient. We discuss the relevant literature regarding incidence and reported gender distribution of inverted takotsubo cardiomyopathy.

[What is the risk of Takotsubo in women?].

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Takotsubo cardiomyopathy preferentially affects postmenopausal women (# 90%). Takotsubo cardiomyopathy mimics an acute coronary syndrome and is defined as a transient left ventricular systolic dysfunction. Diagnosis is based on the criteria of the Mayo Clinic. Stress is found in around 80% of cases: physical stress, emotional stress, medical stress. Physical stress is less found in women. Incidence of Takotsubo cardiomyopathy is 54.9/1,000,000 inhabitants in women versus only 3.6/1,000,000 inhabitants in men. Diagnosis is based on electrocardiogram, biomarkers, echocardiography, coronary angiography and left ventricular angiography, and cardiac magnetic imaging. The complete recovery of left ventricular systolic function must be assessed, allowing to confirm the diagnosis of Takotsubo cardiomyopathy.

Takotsubo cardiomyopathy: Japanese perspective.

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Takotsubo cardiomyopathy is likely induced by stress, and postmenopausal women account for the majority of patients. Several questions about takotsubo cardiomyopathy remain unanswered, including why it is induced by stress, what differentiates the stress that induces this disease from other types of stress that do not induce it, why women are more likely to develop this disease, why the apical part of the heart is affected, and how it is different from the subtype in which the base and middle parts are affected. Although takotsubo cardiomyopathy is uncommon, clinicians should be aware that this disease can develop in wider areas.

Takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy (TC) is characterized by transient left ventricular apical wall motion

abnormalities, chest pain with electrocardiographic changes, and modest myocardial enzymatic release mimicking acute coronary syndrome, but without significant coronary artery disease. TC is an increasingly recognized type of acquired cardiomyopathy occurring commonly after a recent stressful event, in particular emotional stress, and is relatively common in middle-aged and older women. The pathogenetic mechanism remains unknown. Catecholamine surge related to emotional distress seems to play a major role in the pathogenesis of this cardiomyopathy, rendering TC a type of neurocardiological disorder that manifests as acute but reversible heart failure. Clinicians should consider this syndrome in the differential diagnosis of patients presenting with clinical findings suggestive of acute coronary syndrome, especially in postmenopausal women with a recent history of acute emotional or physical stress.

Sex Differences in Stress-Induced (Takotsubo) Cardiomyopathy.

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Takotsubo cardiomyopathy (TC) affects predominantly women. Prior studies have suggested that men might have worse short-term outcomes, but limited data are available regarding long-term outcomes. We hypothesized that men, compared to women, with TC have worse short- and long-term outcomes. A retrospective study of patients diagnosed with TC between 2005 and 2018 in the Veteran Affairs system was performed. Primary outcomes were in-hospital death, 30-day risk of stroke, death, and long-term mortality. A total of 641 patients were included (444 men [69%]; 197 women [31%]). Men had a higher median age (65 vs 60 years; $P < 0.001$), and women were more likely to present with chest pain (68.7% vs 44.1%; $P < 0.001$). Physical triggers were more common in men (68.7% vs 44.1%, $P < 0.001$). Men had a higher in-hospital mortality rate (8.1% vs 1%; $P < 0.001$). On multivariable regression analysis, female sex was an independent predictor for improved in-hospital mortality, compared to men (odds ratio 0.25, 95% confidence interval 0.06-1.10; $P = 0.04$). On 30-day follow-up, no difference occurred in a combined outcome of stroke and death (3.9% vs 1.5%; $P = 0.12$). On long-term follow-up (3.7 ± 3.1 years), female sex was identified as an independent predictor of lower mortality (hazard ratio 0.71, 95% CI 0.51-0.97; $P = 0.032$). Women were more likely to have TC recurrence (3.6% vs 1.1%; $P = 0.04$). In our study with a predominantly male population, men had less-favourable short- and long-term outcomes after TC, compared to those of women.

Cardioprotective Role of Estrogen in Takotsubo Cardiomyopathy.

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Takotsubo cardiomyopathy (TC) is a rare, reversible cause of left ventricular wall motion abnormality (LVWMA) that mimics the presentation of acute myocardial infarction (AMI). TC is usually preceded by an emotional or physical stressor and appears to be more common in postmenopausal women. Various pathophysiological hypotheses of TC have been proposed, but the exact mechanism of action remains elusive. Elevated levels of catecholamines leading to cardiac dysfunction are the most prevalent hypothesis. The protective role of estrogen in the development of cardiomyopathies has been studied extensively. International Takotsubo Diagnostic Criteria (InterTAK) and Mayo clinic diagnostic criteria both have the stipulation stating prevalence of TC is higher in postmenopausal women which hints towards the protective role of estrogen in the development of TC. To review the protective role of estrogen in the mechanism of this novel pathology, we searched Pubmed and Google scholar for the

relevant articles by using keywords such as: "takotsubo cardiomyopathy", "apical ballooning", "broken heart syndrome", "stress cardiomyopathy", "left ventricle wall motion abnormality", "estrogen", "estradiol" and "sex hormones". Our research revealed that although the prevalence of TC is greater in postmenopausal women as compared to men, the prognosis is worse in men. It also revealed the involvement of multiple cellular pathways under the influence of estrogen that could explain the cardioprotective effect of estrogen. Most of the articles found were based on animal studies, thus, there is an emphasis on future human studies. However, we strongly suggest evaluating estrogen levels as part of the initial workup for any patient presenting with signs and symptoms of cardiac pathology.

"Please keep on beating"-Participation in a Creative Workshop Offers Unexpected Benefits to Women With Takotsubo Cardiomyopathy.

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Takotsubo cardiomyopathy (TCM) or "broken heart syndrome" is a rare condition that is more common in women than men, particularly those who are postmenopausal. It mimics a myocardial infarction and psychological factors have been implicated in its etiology as well as being consequences of its presentation. As part of a public engagement project we brought together 8 women (of 12 invited) previously diagnosed with TCM to facilitate a discussion, through participation in a creative workshop-based process, about their illness experience, how they made sense of it, and the meaning it had for them in their lives, and to identify areas of unmet need. Through a range of creative activities we identified that participants had high levels of unmet need in terms of information and psychosocial support. All participants enjoyed the creative process and meeting other people with a diagnosis of TCM. The workshop overall was perceived as empowering. Exploring patient narratives during artist-facilitated workshops is one approach for providing the first steps to addressing unmet need, although the importance of ensuring psychological safety cannot be over-stated.

Takotsubo Cardiomyopathy Associated With Acute Pancreatitis.

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Takotsubo cardiomyopathy is classically associated with emotional stress in middle-aged women. In clinical practice, physical stressors are a more common cause of Takotsubo cardiomyopathy. Here, we present two patients who had acute pancreatitis as a physical stressor that caused Takotsubo cardiomyopathy, and an additional 13 cases identified in the literature. An important clinical feature of these cases is that because metabolic derangements are often encountered, close attention to electrolyte repletion with cardiac monitoring is indicated.

Gender-related differences in takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy (TTC) predominantly occurs in elderly women. Men comprise 10% of the patients, with a similar clinical profile. In contrast to myocardial infarction, age distribution; symptoms,

such as angina; and prehospital delay in TTC are not different between genders. In men, physical stress as a triggering event and shock or cardiac arrest on presentation are more frequent. Gender-related differences in TTC need to be carefully investigated at the clinical and experimental levels to explain the evident gender discrepancy in the prevalence of TTC, to clarify the pathogenetic background, and to develop preventive and therapeutic means against this life-threatening disease.

Pathophysiology of Takotsubo Cardiomyopathy: Reopened Debate.

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Takotsubo cardiomyopathy (TTC), a persistently obscure dysfunctional condition of the left ventricle, is uniquely transient but nevertheless dangerous. It features variable ventricular patterns and is predominant in women. For 30 years, pathophysiologic investigations have progressed only slowly and with inadequate focus. It was initially proposed that sudden-onset spastic obliteration of coronary flow induced myocardial ischemia with residual stunning and thus TTC. Later, it was generally accepted without proof that, in the presence of pain or emotional stress, the dominant mechanism for TTC onset was a catecholamine surge that had a direct, toxic myocardial effect. We think that the manifestations of TTC are more dynamic and complex than can be assumed from catecholamine effects alone. In addition, after reviewing the recent medical literature and considering our own clinical observations, especially on spasm, we theorize that atherosclerotic coronary artery disease modulates and physically opposes obstruction during spasm. This phenomenon may explain the midventricular variant of TTC and the lower incidence of TTC in men. We continue to recommend and perform acetylcholine testing to reproduce TTC and to confirm our theory that coronary spasm is its initial pathophysiologic factor. An improved understanding of TTC is especially important because of the condition's markedly increased incidence during the ongoing COVID-19 pandemic.
